TECHNICAL PROJECT PORTFOLIO / DIGITAL MEDIA TECHNOLOGY

Hostel Discovery & Travel Community Platform

An interactive front-end prototype for bringing hostel discovery, travel guides, and traveller community interfaces into one browsing experience.

PROJECT TYPE	INDIVIDUAL CONTRIBUTION	DEVELOPMENT TOOL
Course project	Individually conceptualised and developed	Cursor (AI-assisted)

01 Project overview

The project originated from an independent trip to Xiamen. During peak travel season, comparing hostel options required searching across several platforms. Accommodation details and traveller experiences were fragmented, making it difficult to compare options efficiently.

The project translated this observation into a product concept: a unified interface through which travellers could discover hostels, browse travel guides, and explore community-oriented content.

Product objective

Explore how a single, structured browsing experience could reduce the effort involved in discovering accommodation information and related travel resources.

02 Product scope

HOSTEL DISCOVERY	MAP-BASED BROWSING
Dynamic hostel cards with region, price-range, and category filters.	Leaflet map interfaces with location markers and OpenStreetMap tiles.
TRAVEL GUIDES	**COMMUNITY INTERFACES**
Guide cards, itinerary details, and route-map interface concepts.	Region and interest groups, organisation information, and activity-map views.

Scope note: this is a front-end prototype. The interface includes example and simulated content; it is not a production booking service.

03 Technical implementation

The prototype uses a lightweight static front-end structure. HTML provides the page and modal structure; CSS controls presentation and responsive breakpoints; JavaScript renders content, applies filters, and coordinates interface interactions.

TECHNOLOGY	ROLE IN THE PROJECT
HTML5	Page structure, navigation, forms, and modal markup.
CSS3	Visual styling, responsive breakpoints, and interface animations.
JavaScript	Dynamic DOM rendering, event handling, filtering, and UI behaviour.
Bootstrap 5.1.3	Responsive layout and modal components.
Leaflet 1.7.1	Interactive map interfaces.
Swiper 8	Homepage image carousel.
Font Awesome 6	Interface iconography.
Cursor	AI-assisted code generation and iterative revision.

04 Development workflow

1	Define the problem and product requirements
2	Translate desired layouts and interactions into natural-language prompts in Cursor
3	Review generated HTML, CSS, and JavaScript
4	Preview the website in a browser and test visible interactions
5	Send observed issues or error messages back to Cursor for targeted revisions
6	Repeat the preview–evaluation–revision cycle

The development approach was prompt-driven: the applicant described requirements and evaluated the generated result, rather than hand-writing all source code from scratch. This distinction is important to an accurate account of the work.

05 Individual contribution

• **Product conception:** identified fragmented hostel information as the problem and proposed a unified discovery experience.

• **Requirement definition:** specified page sections and desired functions, including filtering, maps, guides, and community interfaces.

• **Prompt-driven development:** used natural-language prompts in Cursor to generate and revise the front-end code.

• **Evaluation and iteration:** inspected browser output, tried visible interactions, and used observed issues or error messages to guide further revisions.

06 Limitations and critical reflection

The source is a prototype rather than a deployed full-stack application. It does not include a server-side database, production authentication, persistent user accounts, or a live booking API. Reviews, room types, booking channels, and stories include hard-coded or simulated examples; the membership payment result is simulated in JavaScript.

The source review also identified implementation issues that should be addressed before presenting the website as fully functional: the comments handler refers to HTML elements that are absent; several inline handlers refer to functions not defined in the supplied JavaScript; and the separate *data/hostels.js* file is not imported by the page. A JavaScript syntax check passed, but complete browser-based regression testing remains pending.

The key learning was the difference between describing desired behaviour and understanding how it is implemented. Prompt-driven development made it possible to turn a product idea into an interactive prototype, while exposing gaps in the developer's understanding of program logic, data organisation, and software architecture.

07 Future development

• Consolidate hostel records into a single structured data source.

• Repair unresolved event handlers and validate each visible interaction.

• Separate data and UI logic into maintainable modules.

• Introduce backend services only if persistent accounts, saved items, or genuine bookings become project goals.

• Add automated tests and document browser compatibility.

Evidence and publication

SOURCE REPOSITORY	LIVE DEMO	SCREENSHOTS
To be added after GitHub publication	To be added after deployment and verification	To be added after capturing the actual running website

CV-ready summary: Independently conceptualised and developed an interactive hostel discovery and travel community prototype for a Digital Media Technology course. Translated product requirements into natural-language prompts in Cursor, iteratively generating and refining HTML/CSS/JavaScript interfaces. Evaluated visual output and tested browser interactions, using observed issues and error messages to guide subsequent revisions.

Prepared as an application portfolio draft. Claims are limited to the supplied source code and the developer's description of the development process.